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```
%% Environmental Brain Cube - Modes Analysis
% This script checks how close one mode run was to a chosen condition.
%
% It does these things:
% - reads the modes log file
% - picks one run
% - compares run averages to all conditions
% - makes plots
% - shows simple tables and conclusions
% - makes final match of environment profile
%
```

```
clear; clc; close all;
```

```
%% settings
fileName = "ebc_modes_log.csv";
runChoice = "12";            % "latest" or a run number like "2" (make sure run
number is accurate and quoted"")

%% condition table
% should match Arduino code
profiles = table( ...
    ["NORMAL"; "WARM_HUMID"; "WARM_DRY"; "COLD"], ...
    [20.0; 26.0; 24.0; 4.0], ...
    [24.0; 32.0; 30.0; 12.0], ...
    [30.0; 60.0; 15.0; 35.0], ...
    [50.0; 70.0; 35.0; 65.0], ...
    [995.0; 995.0; 995.0; 995.0], ...
    [1025.0; 1025.0; 1025.0; 1025.0], ...
    'VariableNames',
{'Name','TempMin','TempMax','HumMin','HumMax','PresMin','PresMax'});

%% check file
if ~isfile(fileName)
    error("File not found: " + fileName);
end

%% read file lines
txt = fileread(fileName);
lines = splitlines(string(txt));
lines = strtrim(lines);
lines = lines(lines ~= "");

%% keep only real RUN rows
dataLines = strings(0,1);
```

```
for i = 1:length(lines)
    lineNow = lines(i);

    if startsWith(lineNow, "RUN") && ~startsWith(lineNow, "=== RUN")
        parts = split(lineNow, ",");
        if numel(parts) >= 30
            dataLines(end+1,1) = lineNow; %#ok<SAGROW>
        end
    end
end

if isempty(dataLines)
    error("No valid RUN data rows found.");
end

%% make arrays
n = length(dataLines);

runNum = nan(n,1);
timeSec = nan(n,1);

lightMin = nan(n,1);
lightMax = nan(n,1);
lightState = strings(n,1);

bmeTempMin = nan(n,1);
bmeTempMax = nan(n,1);

bmeHumMin = nan(n,1);
bmeHumMax = nan(n,1);

bmePresMin = nan(n,1);
bmePresMax = nan(n,1);

dsTempMin = nan(n,1);
dsTempMax = nan(n,1);

battMin = nan(n,1);
battMax = nan(n,1);

solarMin = nan(n,1);
solarMax = nan(n,1);

axMin = nan(n,1);
axMax = nan(n,1);

ayMin = nan(n,1);
ayMax = nan(n,1);
```

```
azMin = nan(n,1);
azMax = nan(n,1);

gxMin = nan(n,1);
gxMax = nan(n,1);

gyMin = nan(n,1);
gyMax = nan(n,1);

gzMin = nan(n,1);
gzMax = nan(n,1);

%% pull values from each row
for i = 1:n
    row = split(dataLines(i), ",");
    row = strtrim(row);

    runNum(i) = sscanf(row(1), "RUN%d");
    timeSec(i) = makeNum(row(3));

    lightMin(i) = makeNum(row(4));
    lightMax(i) = makeNum(row(5));
    lightState(i) = row(6);

    bmeTempMin(i) = makeNum(row(7));
    bmeTempMax(i) = makeNum(row(8));

    bmeHumMin(i) = makeNum(row(9));
    bmeHumMax(i) = makeNum(row(10));

    bmePresMin(i) = makeNum(row(11));
    bmePresMax(i) = makeNum(row(12));

    dsTempMin(i) = makeNum(row(13));
    dsTempMax(i) = makeNum(row(14));

    battMin(i) = makeNum(row(15));
    battMax(i) = makeNum(row(16));

    solarMin(i) = makeNum(row(17));
    solarMax(i) = makeNum(row(18));

    axMin(i) = makeNum(row(19));
    axMax(i) = makeNum(row(20));

    ayMin(i) = makeNum(row(21));
    ayMax(i) = makeNum(row(22));

    azMin(i) = makeNum(row(23));
    azMax(i) = makeNum(row(24));
```

```
    gxMin(i) = makeNum(row(25));
    gxMax(i) = makeNum(row(26));

    gyMin(i) = makeNum(row(27));
    gyMax(i) = makeNum(row(28));

    gzMin(i) = makeNum(row(29));
    gzMax(i) = makeNum(row(30));
end

%% make table
T = table(runNum,timeSec,...
    lightMin,lightMax,lightState,...
    bmeTempMin,bmeTempMax,...
    bmeHumMin,bmeHumMax,...
    bmePresMin,bmePresMax,...
    dsTempMin,dsTempMax,...
    battMin,battMax,...
    solarMin,solarMax,...
    axMin,axMax,...
    ayMin,ayMax,...
    azMin,azMax,...
    gxMin,gxMax,...
    gyMin,gyMax,...
    gzMin,gzMax);

%% choose run
allRuns = unique(T.runNum);

if runChoice == "latest"
    runToUse = max(allRuns);
else
    runToUse = double(runChoice);
end

D = T(T.runNum == runToUse, :);
D = sortrows(D, "timeSec");

if isempty(D)
    error("Selected run not found.");
end

fprintf("Using RUN%d\n", runToUse);
```

```
Using RUN12
```

```
fprintf("Intervals logged: %d\n", height(D));
```

```
Intervals logged: 148
```

```
fprintf("Run duration: %.2f s\n", max(D.timeSec));
```

```
Run duration: 300.01 s
```

```
%% midpoint values
D.bmeTempMid = (D.bmeTempMin + D.bmeTempMax) / 2;
D.dsTempMid  = (D.dsTempMin  + D.dsTempMax) / 2;
D.bmeHumMid  = (D.bmeHumMin  + D.bmeHumMax) / 2;
D.bmePresMid = (D.bmePresMin + D.bmePresMax) / 2;

%% average values for fit
avgBmeTemp = mean(D.bmeTempMid, 'omitnan');
avgDsTemp  = mean(D.dsTempMid, 'omitnan');
avgHum     = mean(D.bmeHumMid, 'omitnan');
avgPres    = mean(D.bmePresMid, 'omitnan');

% use BME air temp for matching condition
avgTempForFit = avgBmeTemp;

%% compare to all conditions and rank them
envName = strings(height(profiles),1);

tempPassAll = false(height(profiles),1);
humPassAll = false(height(profiles),1);
presPassAll = false(height(profiles),1);

tempCloseAll = nan(height(profiles),1);
humCloseAll = nan(height(profiles),1);
presCloseAll = nan(height(profiles),1);

tempOffsetAll = nan(height(profiles),1);
humOffsetAll = nan(height(profiles),1);
presOffsetAll = nan(height(profiles),1);

matchCountAll = nan(height(profiles),1);
overallClosenessAll = nan(height(profiles),1);

for k = 1:height(profiles)
    envName(k) = profiles.Name(k);

    [tempPassAll(k), tempCloseAll(k), tempOffsetAll(k)] =
compareToRange(avgTempForFit, profiles.TempMin(k), profiles.TempMax(k));
    [humPassAll(k), humCloseAll(k), humOffsetAll(k)] = compareToRange(avgHum,
profiles.HumMin(k), profiles.HumMax(k));
    [presPassAll(k), presCloseAll(k), presOffsetAll(k)] = compareToRange(avgPres,
profiles.PresMin(k), profiles.PresMax(k));

    matchCountAll(k) = sum([tempPassAll(k), humPassAll(k), presPassAll(k)]);
```

```
    overallClosenessAll(k) = mean([tempCloseAll(k), humCloseAll(k),
presCloseAll(k)], 'omitnan');
end

%% ranked environment table
EnvironmentRanking = table( ...
    envName, ...
    matchCountAll, ...
    overallClosenessAll, ...
    tempPassAll, humPassAll, presPassAll, ...
    tempCloseAll, humCloseAll, presCloseAll, ...
    tempOffsetAll, humOffsetAll, presOffsetAll, ...
    'VariableNames', ...
    {'Environment','MatchCount','OverallClosenessPercent', ...
     'TempPass','HumPass','PresPass', ...
     'TempClosenessPercent','HumClosenessPercent','PresClosenessPercent', ...
     'TempCenterOffsetPercent','HumCenterOffsetPercent','PresCenterOffsetPercent'});

EnvironmentRanking = sortrows(EnvironmentRanking,
{'MatchCount','OverallClosenessPercent'}, {'descend','descend'});

EnvironmentRanking
```

EnvironmentRanking = 4×12 table

...

	Environment	MatchCount	OverallClosenessPercent	TempPass	HumPass
1	"WARM_DRY"	3	61.0482	true	true
2	"WARM_HUMID"	2	47.9852	true	false
3	"NORMAL"	2	38.8025	false	true
4	"COLD"	1	28.9083	false	false

```
%% choose best condition automatically
selectedCondition = EnvironmentRanking.Environment(1);
profileRow = profiles(profiles.Name == selectedCondition, :);

disp("Best matched condition:")
```

```
Best matched condition:
```

```
disp(profileRow)
```

Name	TempMin	TempMax	HumMin	HumMax	PresMin	PresMax
"WARM_DRY"	24	30	15	35	995	1025

```
%% compare to selected condition
[tempPass, tempClose, tempOffset] = compareToRange(avgTempForFit,
profileRow.TempMin, profileRow.TempMax);
```

```
[humPass, humClose, humOffset] = compareToRange(avgHum, profileRow.HumMin,
profileRow.HumMax);
[presPass, presClose, presOffset] = compareToRange(avgPres, profileRow.PresMin,
profileRow.PresMax);

overallCloseness = mean([tempClose, humClose, presClose], 'omitnan');
matchCount = sum([tempPass, humPass, presPass]);

%% comparison table
Parameter = ["Temperature"; "Humidity"; "Pressure"];
RunAverage = [avgTempForFit; avgHum; avgPres];
TargetMin = [profileRow.TempMin; profileRow.HumMin; profileRow.PresMin];
TargetMax = [profileRow.TempMax; profileRow.HumMax; profileRow.PresMax];
Pass = [tempPass; humPass; presPass];
ClosenessPercent = [tempClose; humClose; presClose];
CenterOffsetPercent = [tempOffset; humOffset; presOffset];

ComparisonTable = table(Parameter, RunAverage, TargetMin, TargetMax, ...
    Pass, ClosenessPercent, CenterOffsetPercent);

ComparisonTable
```

ComparisonTable = 3×7 table

...

	Parameter	RunAverage	TargetMin	TargetMax	Pass	ClosenessPercent
1	"Temperature"	27.7169	24	30	true	76.1025
2	"Humidity"	32.9683	15	35	true	20.3172
3	"Pressure"	1.0080e+03	995	1025	true	86.7248

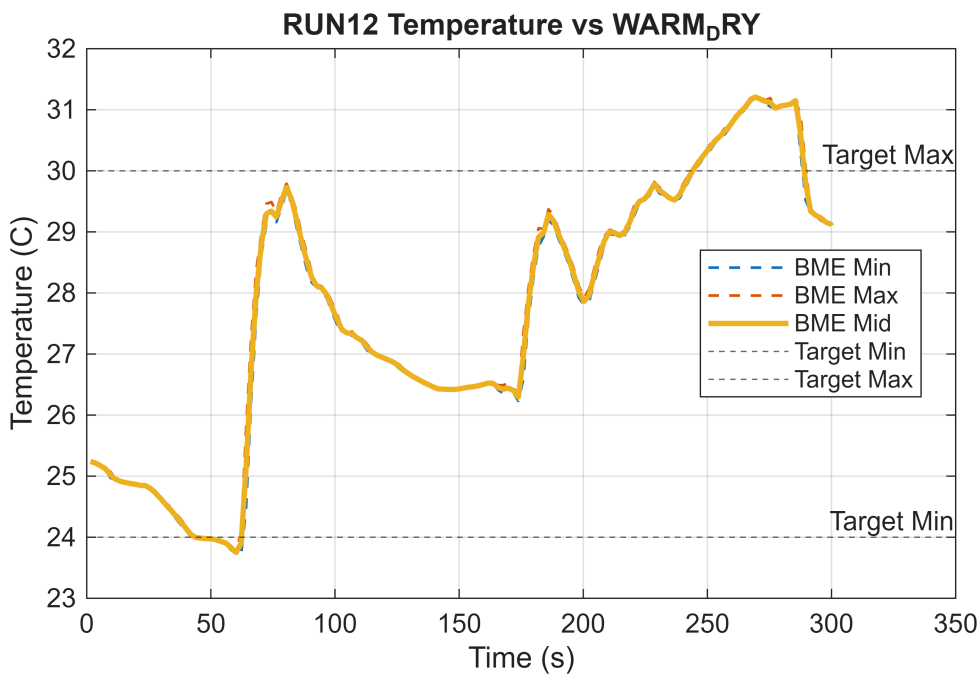
```
%% overall summary table
OverallSummary = table( ...
    string(selectedCondition), ...
    runToUse, ...
    max(D.timeSec), ...
    avgBmeTemp, ...
    avgDsTemp, ...
    avgHum, ...
    avgPres, ...
    matchCount, ...
    overallCloseness, ...
    'VariableNames', ...

{'Condition','Run','DurationSec','AvgBmeTemp','AvgDsTemp','AvgHumidity','AvgPressure
','MatchCount','OverallClosenessPercent'});

OverallSummary
```

OverallSummary = 1×9 table

...

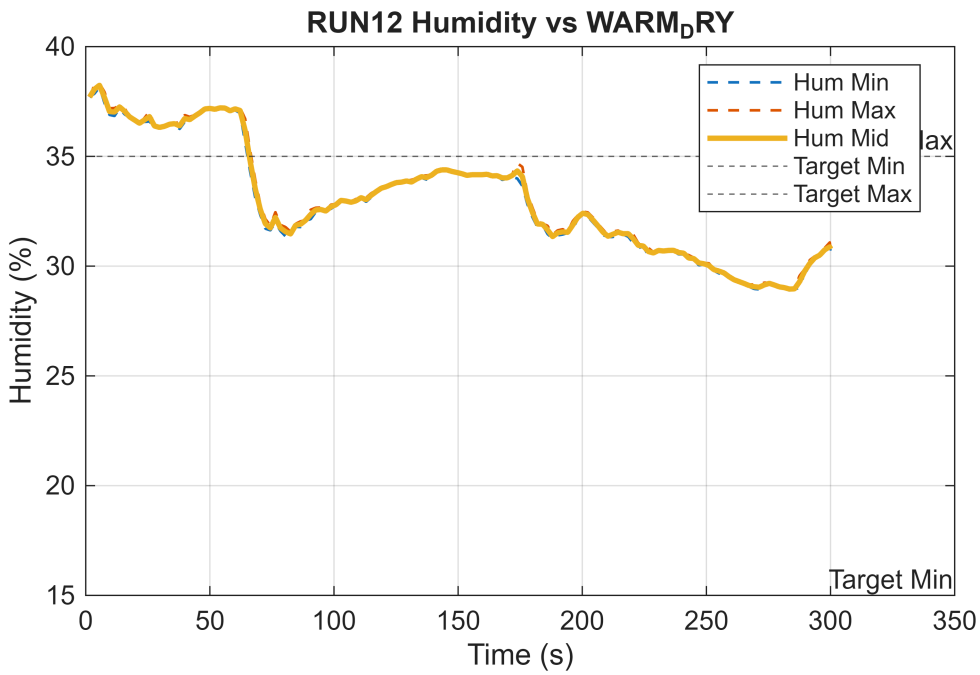
	Condition	Run	DurationSec	AvgBmeTemp	AvgDsTemp	AvgHumidity
1	"WARM_DRY"	12	300.0100	27.7169	20.0572	32.9683

```
%% temperature plot
figure
plot(D.timeSec, D.bmeTempMin, '--', 'LineWidth', 1)
hold on
plot(D.timeSec, D.bmeTempMax, '--', 'LineWidth', 1)
plot(D.timeSec, D.bmeTempMid, 'LineWidth', 2)
yline(profileRow.TempMin, '--', 'Target Min')
yline(profileRow.TempMax, '--', 'Target Max')
grid on
xlabel('Time (s)')
ylabel('Temperature (C)')
title("RUN" + runToUse + " Temperature vs " + selectedCondition)
legend('BME Min','BME Max','BME Mid','Target Min','Target Max','Location','best')
```

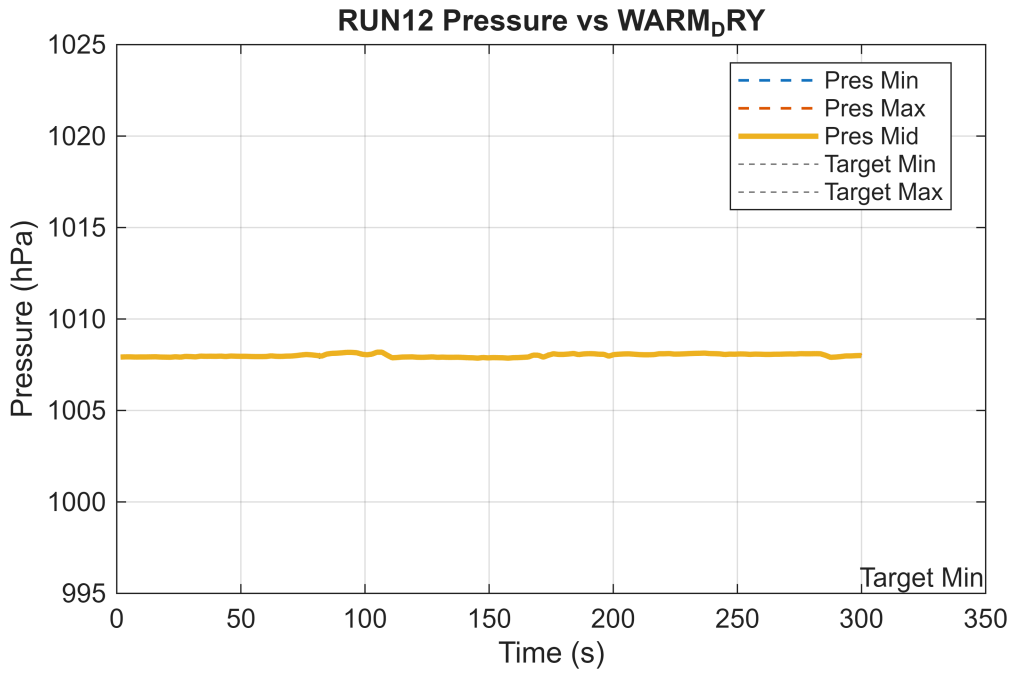



```
%% humidity plot
figure
plot(D.timeSec, D.bmeHumMin, '--', 'LineWidth', 1)
hold on
plot(D.timeSec, D.bmeHumMax, '--', 'LineWidth', 1)
plot(D.timeSec, D.bmeHumMid, 'LineWidth', 2)
yline(profileRow.HumMin, '--', 'Target Min')
yline(profileRow.HumMax, '--', 'Target Max')
grid on
xlabel('Time (s)')
```

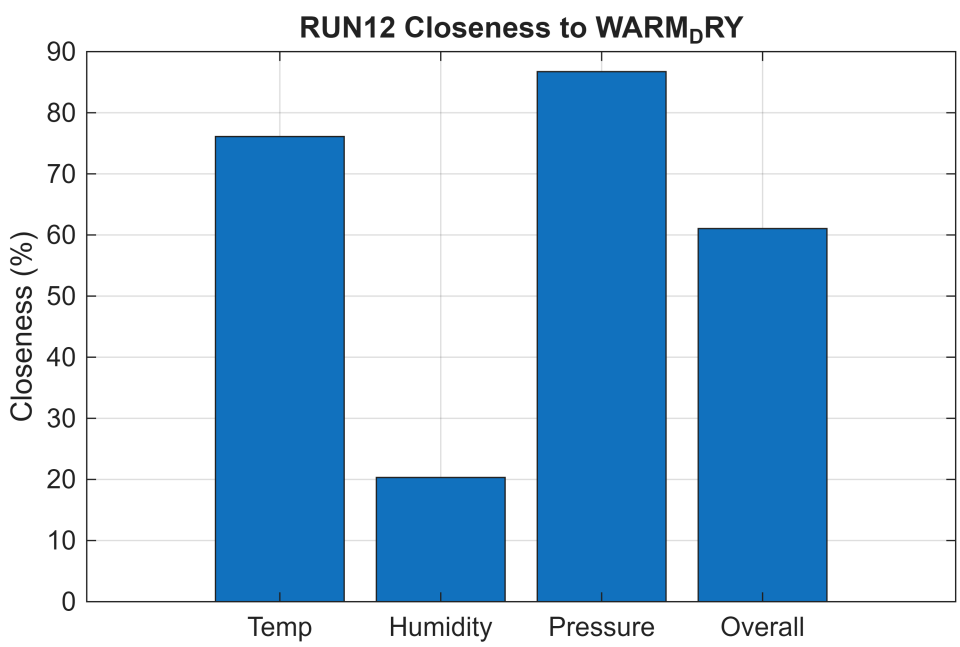
```
ylabel('Humidity (%)')
title("RUN" + runToUse + " Humidity vs " + selectedCondition)
legend('Hum Min','Hum Max','Hum Mid','Target Min','Target Max','Location','best')
```




```
%% pressure plot
figure
plot(D.timeSec, D.bmePresMin, '--', 'LineWidth', 1)
hold on
plot(D.timeSec, D.bmePresMax, '--', 'LineWidth', 1)
plot(D.timeSec, D.bmePresMid, 'LineWidth', 2)
yline(profileRow.PresMin, '--', 'Target Min')
yline(profileRow.PresMax, '--', 'Target Max')
grid on
xlabel('Time (s)')
ylabel('Pressure (hPa)')
title("RUN" + runToUse + " Pressure vs " + selectedCondition)
legend('Pres Min','Pres Max','Pres Mid','Target Min','Target Max','Location','best')
```




```
%% closeness bar chart
figure
bar([tempClose, humClose, presClose, overallCloseness])
set(gca, 'XTickLabel', {'Temp','Humidity','Pressure','Overall'})
ylabel('Closeness (%)')
title("RUN" + runToUse + " Closeness to " + selectedCondition)
grid on
```




```
%% simple conclusions
```

```
fprintf('\n================ CONDITION COMPARISON ================\n');
```

```
================ CONDITION COMPARISON ================
```

```
fprintf('Run analyzed: RUN%d\n', runToUse);
```

```
Run analyzed: RUN12
```

```
fprintf('Best matched condition: %s\n', selectedCondition);
```

```
Best matched condition: WARM_DRY
```

```
fprintf('Run duration: %.2f s\n', max(D.timeSec));
```

```
Run duration: 300.01 s
```

```
fprintf('\nAverage run values used for comparison:\n');
```

```
Average run values used for comparison:
```

```
fprintf('- BME temperature average: %.2f C\n', avgBmeTemp);
```

```
- BME temperature average: 27.72 C
```

```
fprintf('- DS temperature average: %.2f C\n', avgDsTemp);
```

```
- DS temperature average: 20.06 C
```

```
fprintf('- Humidity average: %.2f %%\n', avgHum);
```

```
- Humidity average: 32.97 %
```

```
fprintf('- Pressure average: %.2f hPa\n', avgPres);
```

```
- Pressure average: 1008.01 hPa
```

```
fprintf('\nBest matched condition ranges:\n');
```

```
Best matched condition ranges:
```

```
fprintf('- Temperature: %.1f to %.1f C\n', profileRow.TempMin, profileRow.TempMax);
```

```
- Temperature: 24.0 to 30.0 C
```

```
fprintf('- Humidity: %.1f to %.1f %%\n', profileRow.HumMin, profileRow.HumMax);
```

```
- Humidity: 15.0 to 35.0 %
```

```
fprintf('- Pressure: %.1f to %.1f hPa\n', profileRow.PresMin, profileRow.PresMax);
```

```
- Pressure: 995.0 to 1025.0 hPa
```

```
fprintf('\nParameter results:\n');
```

```
Parameter results:
```

```
fprintf('- Temperature: %s | closeness %.1f %%\n', passText(tempPass), tempClose);
```

```
- Temperature: PASS | closeness 76.1 %
```

```
fprintf('- Humidity: %s | closeness %.1f %%\n', passText(humPass), humClose);
```

```
- Humidity: PASS | closeness 20.3 %
```

```
fprintf('- Pressure: %s | closeness %.1f %%\n', passText(presPass), presClose);
```

```
- Pressure: PASS | closeness 86.7 %
```

```
fprintf('\nOverall:\n');
```

```
Overall:
```

```
fprintf('- Match count: %d/3\n', matchCount);
```

```
- Match count: 3/3
```

```
fprintf('- Overall closeness: %.1f %%\n', overallCloseness);
```

```
- Overall closeness: 61.0 %
```

```
if matchCount == 3
    fprintf('- The run fits the matched condition well overall.\n');
elseif matchCount == 2
    fprintf('- The run is fairly close, but one value is outside range.\n');
elseif matchCount == 1
    fprintf('- The run only weakly matches the matched condition.\n');
else
    fprintf('- The run does not closely match the matched condition.\n');
end
```

```
- The run fits the matched condition well overall.
```

```
fprintf('\nEnvironment ranking from best to worst:\n');
```

```
Environment ranking from best to worst:
```

```
for i = 1:height(EnvironmentRanking)
    fprintf('%d. %s | match %d/3 | closeness %.1f %%\n', ...
        i, EnvironmentRanking.Environment(i), ...
        EnvironmentRanking.MatchCount(i), ...
        EnvironmentRanking.OverallClosenessPercent(i));
end
```

```
1. WARM_DRY | match 3/3 | closeness 61.0 %
2. WARM_HUMID | match 2/3 | closeness 48.0 %
3. NORMAL | match 2/3 | closeness 38.8 %
4. COLD | match 1/3 | closeness 28.9 %
```

```
fprintf('=========================================================\n');
```

```
=====================================================
```

```
function x = makeNum(s)
    s = strtrim(string(s));

    if s == "NA" || s == ""
        x = NaN;
    else
        x = str2double(s);
    end
end

function [isPass, closeness, centerOffsetPercent] = compareToRange(v, vMin, vMax)
    if isnan(v)
        isPass = false;
        closeness = NaN;
        centerOffsetPercent = NaN;
        return
    end

    center = (vMin + vMax) / 2;
    halfRange = (vMax - vMin) / 2;

    if halfRange <= 0
        isPass = false;
        closeness = NaN;
        centerOffsetPercent = NaN;
        return
    end

    centerOffsetPercent = abs(v - center) / halfRange * 100;
    closeness = max(0, 100 - centerOffsetPercent);
    isPass = (v >= vMin) && (v <= vMax);
end

function txt = passText(tf)
    if tf
        txt = 'PASS';
    else
        txt = 'FAIL';
    end
end
```